Physics 143a — Midterm 1 Cheat Sheet (Lecture Notes LN1–LN10)

Labels: (x.y.z) = equation number in the lecture notes; **x.y.z** (bold, e.g. **1.15.4**) = section; **Qx.y.z** (e.g. **Q2.2.3**) = Question; **Ex. x.y.z** (e.g. **Ex. 2.8.1**) = Exercise.
Instruction: Go over Lecture Notes 1 to 5 quickly; then the key concepts in Lecture Notes 6 to 9, including: stationary wavefunction for the time-independent Schrödinger equation: infinite well, finite well, ground and excited states, even and odd states, delta function well, continuity of ψ (always) and ψ' (yes unless infinite potential jump), tunneling, transmission and reflection, lifetime of particle in a potential well.
Very important — LN5 properties of SWFs and completeness: orthonormality $\int \psi_m^* \psi_n \, d^3r = \delta_{mn}$ (1.15.18); completeness, i.e. any $\psi = \sum_n c_n \psi_n(\vec{r}, t)$ (1.15.24), with $c_n = \int \psi_n^* \psi \, d^3r$ (1.15.26). These give $P(E_n) = |c_n|^2$, $\sum |c_n|^2 = 1$, and the time evolution $\psi(x,t) = \sum c_n \psi_n e^{-iE_n t/\hbar}$.
Note: LN10 free particle & Fourier transform might not be tested; worth checking with the professor.
Abbreviations: TDSE = time-dependent Schrödinger equation, $i\hbar\,\partial_t\psi = \hat{H}\psi$; **TISE** = time-independent Schrödinger equation, $\hat{H}\psi = E\psi$; **SWF** = stationary wavefunction, $\psi_E(x,t) = \psi_E(x)e^{-iEt/\hbar}$ (a solution of the TISE times its time factor; a state of definite energy).

KEY RULE: oscillating or exponential? For constant V in a region:
• $E > V$ (classically allowed; the particle can go anywhere): **oscillating** $e^{\pm ikx}$ or $\sin kx, \cos kx$, with $k = \sqrt{2m(E-V)}/\hbar$.
• $E < V$ (forbidden; the particle can't go far): **real exponential** $e^{\pm\kappa x}$, with $\kappa = \sqrt{2m(V-E)}/\hbar$. Keep $e^{+\kappa x}$ for $x \to -\infty$ (left region) and $e^{-\kappa x}$ for $x \to +\infty$ (right region); normalizability kills the other one.
• Inside a *finite* barrier (both ends finite) keep both: $Ce^{\kappa x} + De^{-\kappa x}$.
• ψ is **always** continuous. ψ' is continuous **unless** V jumps by ∞ (δ-function, infinite wall).

LN1 — Blackbody, photons, TDSE

(1.2.3) Planck's law (energy per volume per unit ω):

$$u_P(\omega) = \frac{\hbar\omega^3}{\pi^2 c^3} \frac{1}{e^{\hbar\omega/k_B T} - 1}$$

At low ω it reduces to the classical Rayleigh–Jeans result $\omega^2 k_B T/\pi^2 c^3$ (1.2.7), whose integral over ω diverges (the UV catastrophe). Photon energy $E = h\nu = \hbar\omega$, with $p = h/\lambda = \hbar k$.

Ex. 1.2.4 Peak of $u(\omega)$: $\omega_{\text{peak}} = \dfrac{2.82\,k_B T}{\hbar}$ $\left(\nu_{\text{peak}} = \frac{2.82 k_B T}{h}\right)$. In wavelength, $u(\lambda) = \dfrac{8\pi hc}{\lambda^5} \dfrac{1}{e^{hc/\lambda k_B T} - 1}$ peaks at $\lambda_{\text{peak}} = 0.20 \dfrac{hc}{k_B T}$. Note $\lambda_{\text{peak}} \neq c/\nu_{\text{peak}}$ (it equals $0.567\, c/\nu_{\text{peak}}$), because $u(\lambda)|d\lambda| = u(\omega)|d\omega|$ brings in a Jacobian.

1.5 de Broglie. (1.5.3) $\boxed{\lambda = \dfrac{h}{p}}$ applies to **all particles**. Non-relativistic massive particle (e.g. an electron): $p = \sqrt{2mE}$, so $\lambda = \dfrac{h}{\sqrt{2mE}}$. Photon (massless, fully relativistic): (1.5.2) $E = pc$, so $\lambda = \dfrac{hc}{E}$ (1.5.1). In general $E = \sqrt{p^2c^2 + m_0^2 c^4}$.

(1.6.1) TDSE: $i\hbar \dfrac{\partial \psi(\vec{r},t)}{\partial t} = -\dfrac{\hbar^2}{2m}\nabla^2\psi + V(\vec{r},t)\psi$

LN2 — Probability, current, operators

(1.7.1) $\mathcal{P}(\vec{r},t) = |\psi|^2 = \psi^*\psi$. (1.7.2) Normalization: $\int |\psi|^2 d^3r = 1$; ψ has units of $\mathrm{m}^{-1/2}$ in 1D.

(1.9.5) Probability current: $\vec{J} = -\dfrac{i\hbar}{2m}(\psi^*\nabla\psi - \psi\nabla\psi^*)$

(1.9.6) Continuity equation: $\dfrac{\partial \mathcal{P}}{\partial t} + \nabla \cdot \vec{J} = 0$. (1.9.7) $\dfrac{dP}{dt} = -\oint \vec{J} \cdot d\vec{a}$: the probability in a volume changes only by flow through its boundary. (1.9.8) $\dfrac{dP}{dt} = 0$ over all space, since $J \to 0$ at ∞: a normalized ψ stays normalized.

For a plane wave Ae^{ikx}: $J = \frac{\hbar k}{m}|A|^2 = v|A|^2$. Real ψ (e.g. $\sin kx$) gives $J = 0$.

(1.10.2)–(1.10.8) $E\psi = \hat{H}\psi$; $E \to i\hbar\partial_t$; $\hat{H} = \dfrac{\hat{p}^2}{2m} + \hat{V}$; $\hat{p} \to -i\hbar\nabla$; $\hat{\vec{r}} \to \vec{r}$; $\hat{V} \to V(\vec{r},t)$; $i\hbar\,\partial_t\psi = \hat{H}\psi$.

LN3 — Expectation values, uncertainty

(1.11.1) $P_{AB} = \int_A^B |\psi|^2 dx$. (1.11.2), (1.11.5) $\langle\hat{x}\rangle = \int x|\psi|^2 dx = \int \psi^* \hat{x}\,\psi\, dx$

(1.11.3) $\dfrac{d\langle\hat{x}\rangle}{dt} = \int J\, dx = \dfrac{1}{m}\int \psi^* \left(-i\hbar\partial_x\psi\right) dx$

(1.11.4) $\langle\hat{p}\rangle \equiv m\dfrac{d\langle\hat{x}\rangle}{dt} = \int \psi^* \hat{p}\,\psi\, dx$, with $\hat{p} = -i\hbar\partial_x$. This is equivalent to (1.11.3); it is Ehrenfest's theorem.

(1.11.6) General operator: $\langle\hat{O}\rangle = \int \psi^* \hat{O}(\vec{r}, -i\hbar\nabla)\,\psi\, d^3r$

(1.12.2) $(\Delta O)^2 = \langle(\hat{O} - \langle\hat{O}\rangle)^2\rangle = \langle\hat{O}^2\rangle - \langle\hat{O}\rangle^2$. (1.12.3)

$\boxed{\Delta x\, \Delta p \geq \hbar/2}$

LN4 — Commutators, TISE, stationary states

(1.13.4) $(\hat{x}\hat{p}_x - \hat{p}_x\hat{x})\psi = i\hbar\psi$, i.e. $[\hat{x}, \hat{p}_x] = i\hbar$ (1.13.5). (1.13.7) $[\hat{x}, \hat{p}_x] = [\hat{y}, \hat{p}_y] = [\hat{z}, \hat{p}_z] = i\hbar$. (1.13.8) $[\hat{x}, \hat{p}_y] = [\hat{x}, \hat{y}] = [\hat{p}_x, \hat{p}_y] = 0$. (1.13.9) $[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij}$ and $[\hat{x}_i, \hat{x}_j] = [\hat{p}_i, \hat{p}_j] = 0$.
Operators need not commute ($\hat{A}\hat{B} \neq \hat{B}\hat{A}$) but are always associative: $(\hat{A}\hat{B})\hat{C} = \hat{A}(\hat{B}\hat{C})$. (1.13.10) $[\hat{A}, \hat{B}\hat{C}] = \hat{B}[\hat{A}, \hat{C}] + [\hat{A}, \hat{B}]\hat{C}$. For example, $[\hat{x}, \hat{p}^2] = 2i\hbar\hat{p}$.

Separation of variables (for $V = V(\vec{r})$): (1.14.2) $\psi(\vec{r},t) = \psi(\vec{r})\phi(t)$. Substituting gives (1.14.3); dividing by $\psi\phi$ gives (1.14.4) $i\hbar\dfrac{1}{\phi}\dfrac{d\phi}{dt} = \dfrac{1}{\psi}\left[-\dfrac{\hbar^2}{2m}\nabla^2\psi + V\psi\right] = E$ (a constant, since one side depends only on t and the other only on $\vec{r}$). (1.14.5) $i\hbar\dot{\phi} = E\phi$, so (1.14.6) $\phi(t) = e^{-iEt/\hbar}$. (1.14.7) TISE: $-\dfrac{\hbar^2}{2m}\nabla^2\psi + V\psi = E\psi$, i.e. $\hat{H}\psi = E\psi$ (1.14.8).

(1.14.9) SWF: $\psi_E(\vec{r},t) = \psi_E(\vec{r})\, e^{-iEt/\hbar}$, with E real (1.15.1).

(1.15.2) For a SWF, $\mathcal{P} = |\psi_E(\vec{r})|^2$ is **independent of time**. (1.15.3) $\langle\hat{H}\rangle = E$; (1.15.4) $\langle\hat{H}^2\rangle = E^2$; (1.15.5) $\Delta E = 0$: a state of definite energy. *This holds only for a single SWF, not for a superposition of SWFs with different energies.*

LN5 — Properties of SWFs, completeness, δ-function

1.15.4 The spatial part of a SWF can always be chosen **real** (because ψ^* solves the same TISE). **1.15.5** If $V(-x) = V(x)$,

SWFs can be chosen **even or odd**. In 1D, bound states are nondegenerate (Ex. 2.3.1), so they *must* be even or odd (Ex. 2.3.2).

1.15.6 (1.15.18) Orthonormality: $\int \psi_m^* \psi_n \, d^3r = \delta_{mn}$

1.15.7 (1.15.19) Linearity: $\psi(\vec{r},t) = \sum_n c_n \psi_n(\vec{r}) e^{-iE_n t/\hbar}$ solves the TDSE but is *not* stationary. **1.15.8** (1.15.21) To evolve $\psi(x,0) = \sum c_n \psi_n$, attach $e^{-iE_n t/\hbar}$ to each term.

1.15.9 (1.15.23) $\sum_n |c_n|^2 = 1$. Energy measurement: $P(E_n) = |c_n|^2$ and $\langle E \rangle = \sum |c_n|^2 E_n$ (both time-independent).

1.15.10 (1.15.24) Completeness: any $\psi = \sum_n c_n \psi_n(\vec{r},t)$. (1.15.26) $c_n = \int \psi_n^* \psi \, d^3r$. (1.15.28) $\sum_n \psi_n(\vec{r}',t)^* \psi_n(\vec{r},t) = \delta^3(\vec{r}' - \vec{r})$.

1.16 (1.16.1) $f(x') = \int f(x)\, \delta(x - x')\, dx$; (1.16.3) $\int \delta(x - x')\, dx = 1$; $\delta(ax) = \delta(x)/|a|$.

LN6 — 1D TISE, boundary conditions, infinite well

(2.0.1) $-\dfrac{\hbar^2}{2m}\psi'' + V\psi = E\psi$; (2.0.2) $\psi_E(x,t) = \psi_E(x) e^{-iEt/\hbar}$;

(2.1.1) $\psi'' = \dfrac{2m}{\hbar^2}(V - E)\psi$

(2.1.2) Jump in slope: $\psi'(a^+) - \psi'(a^-) = \dfrac{2m}{\hbar^2} \lim_{\epsilon \to 0} \int_{a-\epsilon}^{a+\epsilon} V\psi \, dx$

2.1 Boundary-condition summary. (1) Finite jump in V: ψ and ψ' are both continuous. (2) Infinite jump (wall or δ): ψ is continuous, ψ' is discontinuous. (3) $\psi \to 0$ at $x = \pm\infty$ (normalizable).

2.2 Infinite well on $[0,a]$. (2.2.1) $\psi = 0$ outside. (2.2.2) $\psi'' + k^2\psi = 0$; (2.2.3) $k = \sqrt{2mE}/\hbar$; $\psi = A \sin kx + B \cos kx$. $\psi(0) = 0$ gives $B = 0$ (a sine), and $\psi(a) = 0$ gives (2.2.5) $k_n = n\pi/a$.

(2.2.6) $\boxed{\psi_n = \sqrt{\tfrac{2}{a}} \sin \tfrac{n\pi x}{a}, \quad E_n = \dfrac{\hbar^2\pi^2 n^2}{2ma^2} \propto \dfrac{n^2}{a^2}}$, $n = 1, 2, 3, \ldots$ $(E_n = n^2 E_1)$

Well on $[0,a]$ is not symmetric about the origin, so the states are not even or odd about 0. For a well on $[-\frac{a}{2}, \frac{a}{2}]$, replace $x \to x + \frac{a}{2}$: $\psi_n = \sqrt{2/a} \sin(\frac{n\pi x}{a} + \frac{n\pi}{2})$, which is $\pm\cos\frac{n\pi x}{a}$ (even) for odd n and $\pm\sin\frac{n\pi x}{a}$ (odd) for even n. Same E_n.

Q2.2.2 At infinite walls only ψ is continuous (ψ' has a kink).

Q2.2.3 The ground state of a symmetric potential is **symmetric**, with no node. **Q2.2.4** ψ_n has $n-1$ **interior nodes**.

Q2.2.5 $\psi_1 + \psi_2$ is not stationary: $|\psi_1(x,t) + \psi_2(x,t)|^2 = \psi_1^2 + \psi_2^2 + 2\psi_1\psi_2 \cos\omega t$, with $\omega = \frac{E_2 - E_1}{\hbar}$, so it oscillates in time.

LN7 — Finite square well, $V = -V_0$ on $[-a,a]$, bound states $-V_0 < E < 0$

Regions I ($x < -a$) and III ($x > a$), where $E < V = 0$: (2.3.5) $\psi'' - \kappa^2\psi = 0$, with $\kappa \equiv \sqrt{-2mE}/\hbar$ ($E < 0$, $V_0 > 0$). (2.3.6) $\psi_I = Be^{\kappa x}$; (2.3.7) $\psi_{III} = Ae^{-\kappa x}$. [The notes print $\hbar^1$ in (2.3.5) and (2.3.8); it should be $\hbar^2$.]

Region II ($|x| \le a$), where $E > V$: (2.3.8) $\psi'' + k^2\psi = 0$, with $k \equiv \sqrt{2m(V_0 + E)}/\hbar$. (2.3.9) $\psi_{II} = C \sin kx + D \cos kx$. *(By convention, Latin k is used inside the well and Greek κ outside.)*

Note that $k^2 + \kappa^2 = 2mV_0/\hbar^2$.

2.3.1 Even states. (2.3.10) $\psi_{II} = D \cos kx$. Continuity at $x = a$: (2.3.11) $D \cos ka = Ae^{-\kappa a}$ (ψ) and (2.3.12) $-kD \sin ka = -\kappa Ae^{-\kappa a}$ (ψ'). Dividing: (2.3.13) $\boxed{k \tan ka = \kappa}$. (2.3.15) Define $z = \dfrac{a\sqrt{2m(V_0 + E)}}{\hbar}$ and $z_0 = \dfrac{a\sqrt{2mV_0}}{\hbar}$. (2.3.16) $\tan z = \sqrt{(z_0/z)^2 - 1}$. Solve graphically. There is **always at least one even state**, however small V_0 is; larger V_0 gives more states.

2.3.2 Odd states. (2.3.17) $\psi_{II} = C \sin kx$. Continuity at $x = -a$: (2.3.18) $C \sin ka = Be^{-\kappa a}$ and (2.3.19) $kC \cos ka = -\kappa Be^{-\kappa a}$. Dividing: (2.3.20) $\boxed{k \cot ka = -\kappa}$, i.e. (2.3.21) $\cot z = -\sqrt{(z_0/z)^2 - 1}$. There may be no odd state: you need $z_0 \ge \pi/2$ (2.3.4). Number of bound states $\approx \lceil 2z_0/\pi \rceil$.

2.3.3 Q2.3.2 A wider or deeper well (larger z_0) has the same or more bound states. If $z_0 \gg 1$: (2.3.22) $z_n \approx n\pi/2$, giving (2.3.23) $E_n \approx -V_0 + \dfrac{\hbar^2\pi^2 n^2}{8ma^2}$, which is the infinite well (2.2.6) with $a \to 2a$, measured from $-V_0$.

2.3.4 Q2.3.3 The ground-state energy is never at the bottom: $\psi'' \neq 0$, and $\Delta x \neq 0$ implies $\Delta p \neq 0$. **Q2.3.4** The finite-well E_n (measured from the bottom) are **lower** than the infinite-well ones, because ψ leaks out and curves less. **Q2.3.1** ψ decays exponentially outside the well. The ground state is even with no nodes, nodes increase by one per level, and there are finitely many bound states.

LN8 — δ-function well; reflection and transmission

2.4 (2.4.1) $V = -\alpha\delta(x)$ with $\alpha > 0$. (2.4.2) $\psi'' - \kappa^2\psi = 0$, $\kappa = \sqrt{-2mE}/\hbar$. (2.4.3) $\psi_I = Be^{\kappa x}$ ($x < 0$); (2.4.4) $\psi_{II} = Ae^{-\kappa x}$ ($x > 0$). (2.4.5) ψ continuous: $A = B$. (2.4.6) ψ' jumps: $\psi'_{II}(0) - \psi'_I(0) = -\dfrac{2m\alpha}{\hbar^2}\psi(0)$, which gives (2.4.7) $\kappa A + \kappa B = \frac{2m}{\hbar^2}\alpha A$.

(2.4.8) $\kappa = \dfrac{m\alpha}{\hbar^2}$, (2.4.9) $\boxed{E = -\dfrac{m\alpha^2}{2\hbar^2}}$, $\psi = \sqrt{\kappa}\, e^{-\kappa|x|}$. There is **exactly one** bound state. *For a δ barrier $+V_0\delta(x)$: $\psi'(0^+) - \psi'(0^-) = +\frac{2mV_0}{\hbar^2}\psi(0)$; with $\beta = \frac{mV_0}{\hbar^2 k}$, $F/A = \frac{1}{1+i\beta}$, $B/A = \frac{-i\beta}{1+i\beta}$, $T = \frac{1}{1 + mV_0^2/2\hbar^2 E}$ (PS4 #1).*

2.5 (2.5.1) $R = \dfrac{J_{\text{ref}}}{J_{\text{inc}}}$, $T = \dfrac{J_{\text{tra}}}{J_{\text{inc}}}$, with $J = \mathcal{P}\, v = v|A|^2$. Use currents, not just $|\psi|^2$, when the speeds differ. Particle with $E > 0$ passing over a well $-V_0$ on $[-a,a]$: (2.5.2) $\psi_I = Ae^{ikx} + Be^{-ikx}$; (2.5.3) $\psi_{II} = C \sin \kappa x + D \cos \kappa x$; (2.5.4) $\psi_{III} = Fe^{ikx}$ (no wave coming back from the right). (2.5.5) $k = \sqrt{2mE}/\hbar$, $\kappa = \sqrt{2m(E + V_0)}/\hbar$ [the notes print both as κ]. Continuity of ψ and ψ' at $\pm a$ gives (2.5.6)–(2.5.9), which solve to (2.5.10) $F = \dfrac{e^{-2ika}A}{\cos 2\kappa a - i\frac{k^2 + \kappa^2}{2k\kappa}\sin 2\kappa a}$ and (2.5.11) $B = i\dfrac{\sin 2\kappa a}{2k\kappa}(\kappa^2 - k^2)F$.

(2.5.12) $T = \left|\dfrac{F}{A}\right|^2 = \dfrac{1}{1 + \frac{V_0^2}{4E(E+V_0)} \sin^2\left(\frac{2a}{\hbar}\sqrt{2m(E+V_0)}\right)}$; (2.5.13) $R = 1 - T$. T oscillates with E (Fig. 2.5.2). $T = 1$ when $2\kappa a = n\pi$, i.e. at the infinite-well energies (the Ramsauer–Townsend effect, Q2.5.3).

Q2.5.4 Where V is lower, the particle has more kinetic energy: higher momentum, oscillates faster (shorter λ), moves faster, so it is less likely to be found there and the **amplitude is smaller**.

LN9 — Step, tunneling, lifetime, harmonic oscillator

2.6 Step ($V = V_0$ for $x > 0$), with $E > V_0$: (2.6.1) $\psi_I = Ae^{ikx} + Be^{-ikx}$, $\psi_{II} = Ce^{i\kappa x}$; (2.6.2) $k = \sqrt{2mE}/\hbar$, $\kappa = \sqrt{2m(E - V_0)}/\hbar$. (2.6.3) $A + B = C$ and $ik(A - B) = i\kappa C$, giving (2.6.4) $B = \frac{k - \kappa}{k + \kappa}A$ and $C = \frac{2k}{k+\kappa}A$. (2.6.5) $J_{\text{inc}} = \frac{\hbar k}{m}|A|^2$, $J_{\text{ref}} = \frac{\hbar k}{m}|B|^2$, $J_{\text{tra}} = \frac{\hbar\kappa}{m}|C|^2$ (i.e. $J = v|\text{amp}|^2$).

(2.6.6) $R = \left(\frac{k-\kappa}{k+\kappa}\right)^2 = \left(\frac{\sqrt{E}-\sqrt{E-V_0}}{\sqrt{E}+\sqrt{E-V_0}}\right)^2$ and $T = \frac{\kappa}{k}\left|\frac{C}{A}\right|^2 = \frac{4k\kappa}{(k+\kappa)^2}$. (2.6.7) $R+T=1$. For $E<V_0$: $\psi_{II} = Ce^{-\kappa x}$ is real, so $J_{\text{tra}} = 0$ and $R = 1$ (Q2.6.2). Classically $R = 1$ for $E < V_0$ and $R = 0$ for $E > V_0$.

2.8 Tunneling through a barrier V_0 on $[-a, a]$, with $0 < E < V_0$: (2.8.1) $\psi_I = Ae^{ikx} + Be^{-ikx}$, $\psi_{II} = Ce^{\kappa x} + De^{-\kappa x}$, $\psi_{III} = Fe^{ikx}$; (2.8.2) $k = \frac{\sqrt{2mE}}{\hbar}$, $\kappa = \frac{\sqrt{2m(V_0-E)}}{\hbar}$. (2.8.3) Solving for B, C, D, F in terms of A gives $F = \frac{4ik\kappa\, e^{2a(\kappa-ik)}}{e^{4a\kappa}(k+i\kappa)^2 - (k-i\kappa)^2}A$. Any E is allowed (not quantized).

(2.8.4) $T = \left|\frac{F}{A}\right|^2 = \frac{(2k\kappa)^2}{(k^2+\kappa^2)^2\sinh^2(2\kappa a) + (2k\kappa)^2}$. For $V_0 \gg E$: (2.8.5) $T \approx \left(\frac{4k\kappa}{k^2+\kappa^2}\right)^2 e^{-2\kappa(2a)}$, so (2.8.6) $\ln T \approx -2\kappa(2a)$ and $T \approx e^{-2\kappa\cdot 2a}$ (2.8.7).

General barrier shape: (2.8.10) $\boxed{T \approx e^{-2\gamma}}$, (2.8.11) $\gamma = \int_a^b \frac{\sqrt{2m(V(x)-E)}}{\hbar}dx$, integrated over the forbidden region.

Ex. 2.8.1 Lifetime of a particle in a well ($V=\infty$ at $x<0$, well $0<x<a$, barrier V_0 on $a<x<b$):
speed $v = \sqrt{2E/m}$; tunneling probability per hit $T = e^{-2\gamma}$, with $\gamma = \frac{\sqrt{2m(V_0-E)}}{\hbar}(b-a)$;
hits needed $N \approx 1/T = e^{2\gamma}$; time between hits on the barrier $\Delta t = 2a/v$;

$$\boxed{\tau \approx N\Delta t = \frac{2a}{v}e^{2\gamma} = \frac{2a}{\sqrt{2E/m}}\, e^{\frac{2}{\hbar}\sqrt{2m(V_0-E)}\,(b-a)}}$$

Two leaky walls (a well of width w between two barriers): $\Delta t = w/v$, because every crossing ends at a barrier.

2.7 Harmonic oscillator. (2.7.1) $V = \frac{1}{2}m\omega^2x^2$. (2.7.4) $\xi = \sqrt{m\omega/\hbar}\,x$ and (2.7.6) $K = 2E/\hbar\omega$ turn the TISE into (2.7.5) $\frac{d^2\psi}{d\xi^2} = (\xi^2 - K)\psi$. For large ξ, $\psi \sim e^{-\xi^2/2}$, so try the ansatz (2.7.9) $\psi = H(\xi)e^{-\xi^2/2}$. This gives the Hermite equation (2.7.12) $H'' - 2\xi H' + (K-1)H = 0$ (here H is *not* the Hamiltonian).

Power series (2.7.13) $H = \sum a_n\xi^n$ leads to (2.7.15) $a_{n+2} = \frac{2n+1-K}{(n+2)(n+1)}a_n$. Starting from a_0 gives even solutions; from a_1, odd solutions. An infinite series behaves like e^{ξ^2} and is not normalizable, so the series must **truncate**: $2n+1-K=0$, giving (2.7.17) $\boxed{E_n = (n+\tfrac{1}{2})\hbar\omega}$.

(2.7.18), (2.7.19) $E_0 = \frac{1}{2}\hbar\omega$, $\psi_0 = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega x^2}{2\hbar}}$. (2.7.20), (2.7.21) $E_1 = \frac{3}{2}\hbar\omega$, $\psi_1 = \left(\frac{4}{\pi}\left(\frac{m\omega}{\hbar}\right)^3\right)^{1/4} x\, e^{-\frac{m\omega x^2}{2\hbar}}$ [the notes label (2.7.20) "E_0"; it should be E_1].

(2.7.22) $\psi_n(\xi) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\frac{1}{\sqrt{2^n n!}}H_n(\xi)e^{-\xi^2/2}$; (2.7.23) Rodrigues formula $H_n = (-1)^n e^{\xi^2}\frac{d^n}{d\xi^n}e^{-\xi^2}$. $H_0 = 1$, $H_1 = 2\xi$, $H_2 = 4\xi^2 - 2$, $H_3 = 8\xi^3 - 12\xi$. In the ground state, $\Delta x\Delta p = \hbar/2$ (the minimum).

Q2.7.2 For large n (the classical limit), $|\psi_n|^2$ peaks near the classical **turning points**, because the particle moves slowest there. This is the opposite of the ground state, which peaks at the center.

Ex. 2.7.1 Gaussian integrals ($a>0$; integrals over $(-\infty,\infty)$ unless shown):

(1) $\int_{-\infty}^{\infty} e^{-ax^2}dx = \sqrt{\frac{\pi}{a}}$, $\int_0^{\infty} e^{-ax^2}dx = \frac{1}{2}\sqrt{\frac{\pi}{a}}$

(2) $\int x^{2n+1}e^{-ax^2}dx = 0$

(3) $\int x^{2n}e^{-ax^2}dx = (-1)^n\frac{d^n}{da^n}\sqrt{\frac{\pi}{a}}$
$= \frac{(2n-1)!!}{(2a)^n}\sqrt{\frac{\pi}{a}} = \frac{(2n)!}{n!\,(4a)^n}\sqrt{\frac{\pi}{a}}$
e.g. $n=1$: $\frac{1}{2a}\sqrt{\frac{\pi}{a}}$; $n=2$: $\frac{3}{4a^2}\sqrt{\frac{\pi}{a}}$

(4) $\int e^{-ax^2+bx}dx = \sqrt{\frac{\pi}{a}}\,e^{b^2/4a}$

(5) $\int x^n e^{-ax^2+bx}dx = \frac{d^n}{db^n}\left(\sqrt{\frac{\pi}{a}}\,e^{b^2/4a}\right)$

LN10 — Free particle, Fourier transform (may not be tested — check with the professor)

(2.10.3) $\psi'' + k^2\psi = 0$, $k = \sqrt{2mE}/\hbar$. (2.10.4) $\psi = Ae^{i(kx-\omega t)} + Be^{-i(kx+\omega t)} = Ae^{ik(x-v_pt)} + Be^{-ik(x+v_pt)}$. (2.10.5) $\omega = \frac{E}{\hbar} = \frac{\hbar k^2}{2m}$. (2.10.6) $v_p = \frac{\omega}{k} = \frac{\hbar k}{2m} = \frac{1}{2}v_{cl}$. (2.10.7) $\psi = Ae^{ikx - i\hbar k^2t/2m}$ is not normalizable.

(2.10.9) $\psi(x,t) = \frac{1}{\sqrt{2\pi}}\int dk\,\phi(k)e^{ikx-i\omega(k)t}$; (2.10.10) $\phi(k,t) = \phi(k)e^{-i\omega(k)t}$: each $\phi(k)$ is a state of definite energy $E = \hbar\omega = \frac{\hbar^2k^2}{2m}$, so it evolves like a stationary state (just a phase); (2.10.12) $\psi(x) = \frac{1}{\sqrt{2\pi}}\int dk\,\phi(k)e^{ikx}$; (2.10.14) $\phi(k) = \frac{1}{\sqrt{2\pi}}\int dx\,\psi(x)e^{-ikx}$; (2.10.16) $\delta(x-x') = \frac{1}{2\pi}\int dk\,e^{-ik(x-x')}$.

(2.10.17) Expand $\omega(k) = \omega_0 + \omega'(k_0)(k-k_0) + \frac{1}{2}\omega''(k_0)(k-k_0)^2 + \ldots$ (2.10.18) Then $kx - \omega t = k_0(x-v_pt) + (k-k_0)(x - v_gt) - \frac{1}{2}\alpha^2(k-k_0)^2t$, with (2.10.19) $v_p = \omega_0/k_0$ (phase velocity), (2.10.20) $v_g = \frac{d\omega}{dk}\big|_{k_0}$ (group velocity), and (2.10.21) $\alpha^2 = \frac{d^2\omega}{dk^2}\big|_{k_0}$. (2.10.22) For a free particle, $v_p = \frac{\hbar k_0}{2m}$ and $v_g = \frac{\hbar k_0}{m} = v_{cl}$: the packet moves at v_g.

2.10.1 Rectangular amplitude $\phi(k) = 1/\sqrt{\Delta k}$ for $|k| < \Delta k/2$: (2.10.24) $\psi(x,0) = \frac{1}{\sqrt{2\pi\Delta k}}\frac{2}{x}\sin\frac{\Delta k\,x}{2}$, a packet concentrated at the origin that falls off at large x. (2.10.25) $\Delta x = 4\pi/\Delta k$ (nodes at $\pm 2\pi/\Delta k$); (2.10.26) $\Delta x\Delta p = 4\pi\hbar$. (2.10.27) For $\Delta k\, x \ll 1$, $\psi \approx \sqrt{\Delta k/2\pi}$, i.e. constant: precise momentum means uncertain position.

2.10.2 Rectangular pulse $\psi(x) = 1/\sqrt{\Delta x}$ for $|x| < \Delta x/2$: (2.10.28) $\phi(k) = \frac{1}{\sqrt{2\pi\Delta x}}\frac{2}{k}\sin\frac{k\Delta x}{2}$, concentrated at $k=0$ — the exact counterpart of 2.10.1. (2.10.29) $\Delta k = 4\pi/\Delta x$.

Gaussian packet (PS5): $\psi = (\pi a^2)^{-1/4}e^{-x^2/2a^2}e^{ik_0x}$, $a(t) = a\sqrt{1+\hbar^2t^2/m^2a^4}$, $\Delta x\Delta p = \frac{\hbar}{2}\sqrt{1+\hbar^2t^2/m^2a^4}$.